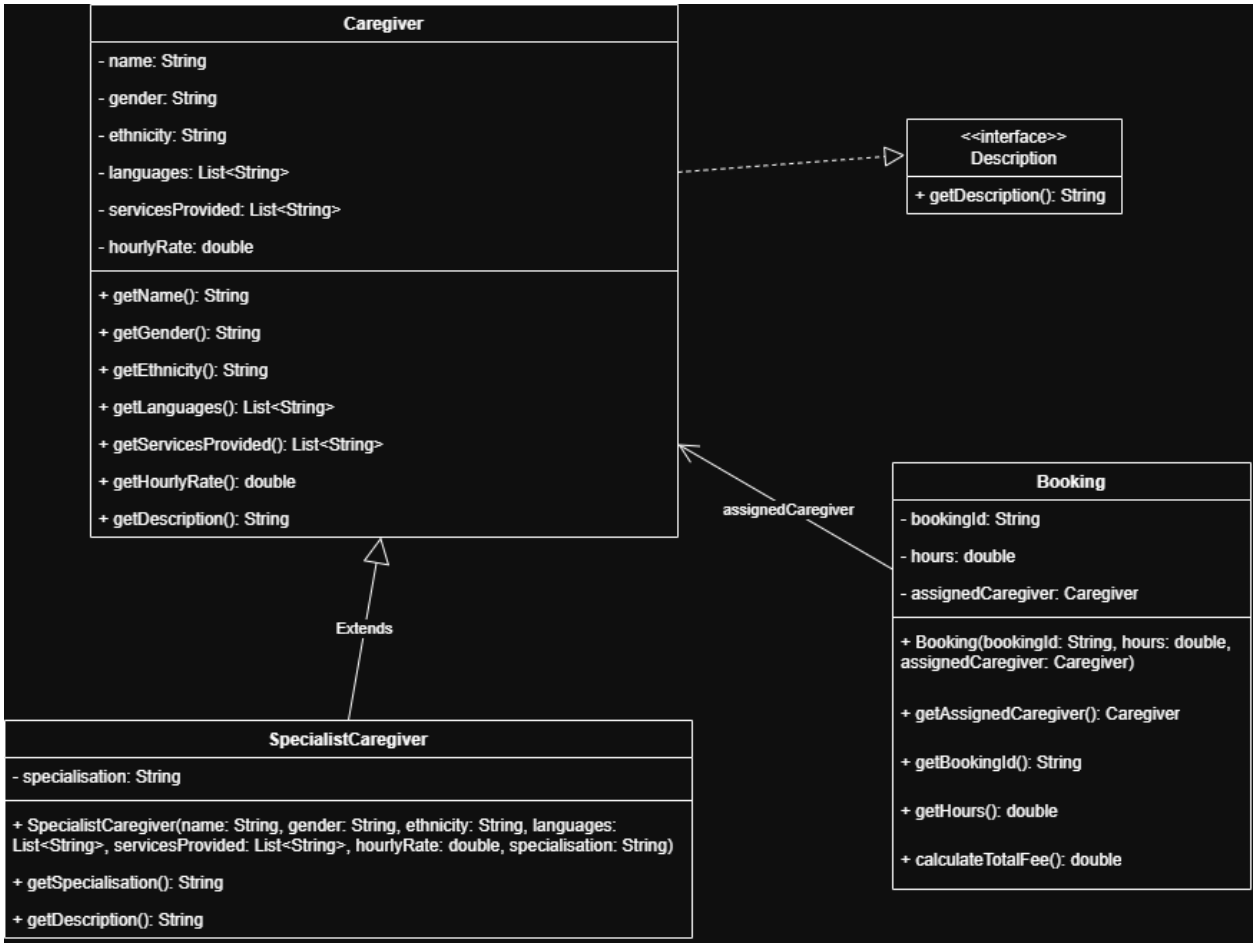
The problem
The Care Service Booking System supports elderly people who have limited mobility or need assistance with activities of daily living (ADLs). It also helps family members who cannot provide care during working hours or who live separately. The system helps users find a suitable caregiver for the required tasks and clearly displays the service fees and duration before booking.

Why Java?
Java's object-oriented features allow the system to represent caregivers, specialist caregivers and bookings as separate classes. Inheritance lets SpecialistCaregiver reuse common caregiver details while overriding getDescription() to provide a specialised description. Java's static type checking also helps detect incompatible data types during compilation.

Alternative and why it is less suitable
JavaScript is a possible alternative. However, for this project's requirement to demonstrate explicitly typed classes, interfaces and inheritance, plain JavaScript is less suitable because it does not provide Java-style interfaces or compile-time static type checking. Additional tooling, such as TypeScript, would be needed to provide comparable checks.

Classes Diagram



Objects Diagram

